

UNIT-V - Probabilistic Reasoning / Inference.

- Bayesian Network.

* Basics of probability.

- Random variables

Rain $R \in \{0, 1\}$

Sunshine $S \in \{0, 1\}$.

} The values might be b/w 0 & 1 also.

S	R	$P(S=s, R=r)$
0	0	0.20
0	1	0.08
1	0	0.70
1	1	0.02
		sum = 1.00.

prob. (no rain, no sun).

} distribution.

→ Joint Distribution

↓ Any prob distⁿ, the total sum = 1.00.

* In random variables, the outcome is probability, where as in logic variable the outcome is 0/1.

* Marginal Distribution

* There are 2 variables & out of that, we take 1.

$P(S) =$

S	$P(S=s)$
0	0.28
1	0.72

→ marginalising / ignoring 1 set of values (R). & summing over other variable values.

$$P(S=0) = P(S=0, R=0) + P(S=0, R=1)$$

$$\Rightarrow P(S=0) = 0.20 + 0.08 = 0.28$$

$$P(R=1) = P(S=0, R=1) + P(S=1, R=1)$$

$$= 0.08 + 0.02 = 0.10$$

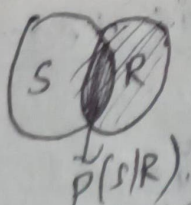
* Conditional Distribution.

$P(S|R=1) =$

S	$P(S=s R=1)$
0	0.08
1	0.02
0.10	

X

$$P(S|R) = \frac{P(S \cap R)}{P(R)}$$



$$P(S=0|R=1) = \frac{P(S=0, R=1)}{P(R=1)} = \frac{0.08}{0.10} = 0.8$$

$$P(S=1|R=1) = \frac{P(S=1, R=1)}{P(R=1)} = \frac{0.02}{0.10} = 0.2$$

S	$P(S=S R=1) = P(S R=1)$
0	0.8
1	0.2
	1.00

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

conditional

joint

marginal

* we can get 1 distn using other 2.

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \rightarrow \text{Bayes Theorem}$$

It is used to do machine learning reasoning.

ex: for a doctor $\rightarrow P(\text{cause}|\text{effect}) = \frac{P(\text{eff}|\text{cause}) \cdot P(\text{cause})}{P(\text{effect})}$

it can be found from data.

ex: Rain $\in \{0, 1\}$

Grass is wet $\in \{0, 1\}$

$P(\text{Rain} | \text{Grass is wet}) \rightarrow$ reasoning (b/c Grass is wet b/c of any other reason)...

$P(\text{Grass is wet} | \text{Rain}) \rightarrow$ data

$$P(\text{Rain} | \text{Grass is wet}) = \frac{P(\text{Grass is wet} | \text{Rain}) \cdot P(\text{Rain})}{P(\text{Grass is wet})}$$

$$P(\text{Rain}) = 0.3 \Rightarrow \text{It rains 30\% of the days.}$$

$$P(G=1 | R=1) = 0.9 \Rightarrow \text{If it rains grass is wet 90\% of time.}$$

$$P(G=1, R=0) = 0.1 \Rightarrow \text{Even if not rains, grass is wet 10\% of time.}$$

$$P(R=1 | G=1) = \frac{P(G=1 | R=1) \cdot P(R=1)}{P(G=1)}$$

$$= \frac{0.9 \times 0.3}{P(G=1, R=0) + P(G=1, R=1)} = \frac{0.9 \times 0.3}{0.9 + 0.1}$$

$$\begin{aligned} P(G=1) &= P(G=1, R=0) + P(G=1, R=1) \\ &= P(G=1 | R=0) \cdot P(R=0) + P(G=1 | R=1) \cdot P(R=1) \\ &= 0.1 \times 0.7 + 0.9 \times 0.3 \\ &= 0.07 + 0.27 = \boxed{0.34} \end{aligned}$$

$$\therefore P(R=1 | G=1) = \frac{0.9 \times 0.3}{0.34} = \boxed{0.794} //$$

If grass is wet, 80\% prob of Raining.

* For spam detection also, it is used, (based on words)
i.e. $P(\text{spam} | \text{lottery}) = \frac{P(\text{lottery} | \text{spam}) \cdot P(\text{spam})}{P(\text{lottery})}$

* Independence

* If A, B are independent

$$\boxed{P(A \cap B) = P(A) \cdot P(B)}$$

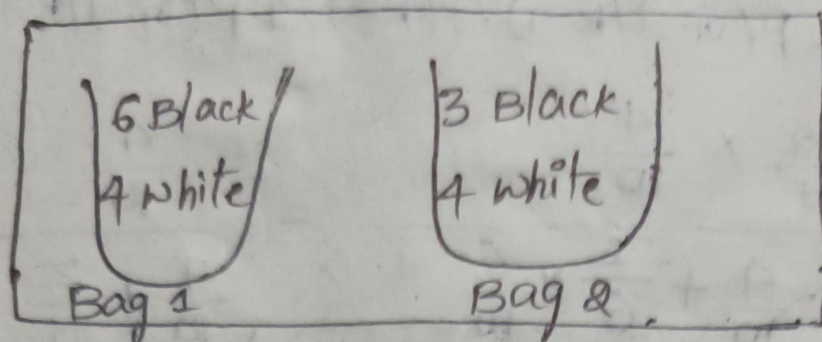
* Conditional Independence

$$\boxed{P(A, B | C) = P(A | C) \cdot P(B | C)}$$

* If 3rd factor is known, then A, B b/m independent.

* Many Q's we are asking are in probabilistic Nature.

Continue
Ex: →



$$P(\text{Black} | \text{Bag 1}) = \frac{6}{10} = 0.6$$

$$Q: P(\text{Bag 1} | \text{Black}) = \frac{P(\text{Black} | \text{Bag 1}) \cdot P(\text{Bag 1})}{P(\text{Black})}$$

$$P(\text{Black} | \text{Bag 1}) = 6/10 = 0.6$$

$$P(\text{Bag 1}) = 1/2 = 0.5$$

$$P(\text{Black}) = P(\text{Black}, \text{Bag 1}) + P(\text{Black}, \text{Bag 2})$$

$$= P(\text{Black} | \text{Bag 1}) \cdot P(\text{Bag 1}) + P(\text{Black} | \text{Bag 2}) \cdot P(\text{Bag 2})$$

$$= \left(\frac{6}{10}\right)(0.5) + \left(\frac{3}{7}\right)(0.5)$$

$$= 0.51$$

$$\therefore P(\text{Bag 1} | \text{Black}) = \frac{0.6 \times 0.5}{0.51} = \boxed{0.588}$$

Ex: Rain, Grass wet

Days	1	2	3	4	5	6	7	8	9	10
R	1	0	1	0	0	1	0	0	1	0
G	1	0	1	1	0	1	0	1	1	0

$$\rightarrow P(R=1 | G=1) = \frac{4}{6} = \frac{2}{3} //$$

$$\rightarrow P(R=1 | G=1) = \frac{P(G=1 | R=1) \cdot P(R=1)}{P(G=1)}$$

$$P(G=1) = P(G=1, R=1) + P(G=1, R=0)$$

$$P(G=1) = P(G=1 | R=1) \cdot P(R=1) + P(G=1 | R=0) \cdot P(R=0)$$

$$\therefore P(R=1 | G=1) = \frac{P(G=1 | R=1) \cdot P(R=1)}{P(G=1 | R=1) \cdot P(R=1) + P(G=1 | R=0) \cdot P(R=0)}$$

$$= \frac{\frac{4}{4} \times \frac{4}{10}}{\frac{4}{4} \cdot \frac{4}{10} + \frac{2}{6} \times \frac{6}{10}} = \frac{\frac{4}{10}}{\frac{4}{10} + \frac{2}{10}} = \frac{4}{6} //$$

* Independence & conditional Independence

Independence

$$P(X, Y) = P(X) \cdot P(Y) \quad X, Y \text{ are Independent}$$

Ex: $P(\text{Red shirt, wife is down}) = P(\text{red shirt}) * P(\text{wife is down})$

$$P(\text{Grass wet, Rain}) \neq P(\text{Grass wet}) * P(\text{Rain})$$

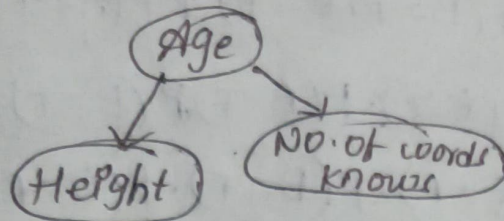
* Conditional Independence

X: Height of a child

Y: No. of words the child knows.

If ht is more, he knows more words $\rightarrow$ Is it true?

* Actually, the x, y are dependent on 3rd variable
 i.e. on - Age of the child.
 z : Age of the child.



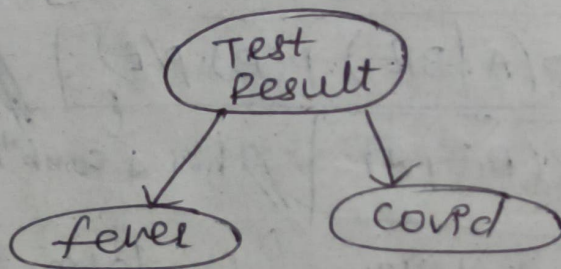
Conditional Independence

$$P(x, y|z) = P(x|z) \cdot P(y|z)$$

$$\Rightarrow (x \perp\!\!\!\perp y) | z \Leftrightarrow P(x, y|z) = P(x|z) \cdot P(y|z)$$

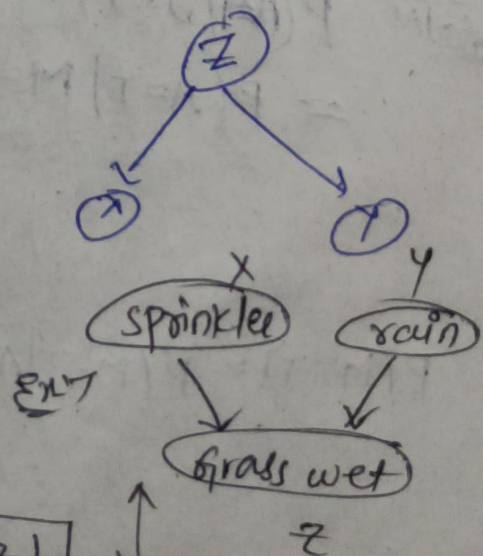
Given the knowledge that z occurs, knowledge of whether x occurs doesn't provide additional information on likelihood of y occurs, & vice versa where $\perp\!\!\!\perp$ represents independence.

Ex 1



↓ Another form of conditional independence:→

$$\begin{aligned}
 P(x|y, z) &= \frac{P(x, y, z)}{P(y, z)} \\
 &= \frac{P(x|y, z) \cdot P(y, z)}{P(y, z) \cdot P(z)} \\
 &= \frac{P(x|z) \cdot P(y|z) \cdot P(z)}{P(y, z) \cdot P(z)} \\
 &= P(x|z)
 \end{aligned}$$



$$x \perp\!\!\!\perp y | z \Rightarrow P(x|y, z) = P(x|z)$$

* Independence

Continue

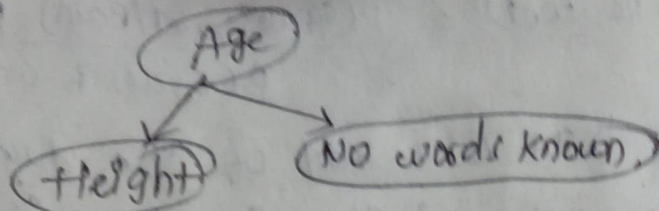
* If A, B are independent

$$P(A, B) = P(A) \cdot P(B)$$

* conditional Independence

$$P(A, B|C) = P(A|C) \cdot P(B|C)$$

* If 3rd factor is known, then A, B b/m independent.



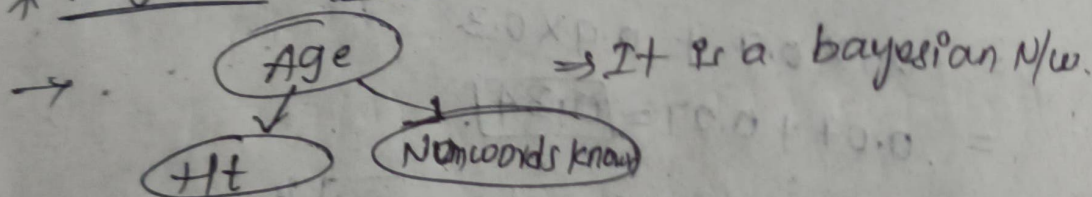
$$\star \boxed{P(A|B, C) = P(A|C)}$$

$$\star P(A|B, C) = \frac{P(A|B, C) \cdot P(B|C) \cdot P(C)}{P(B|C) \cdot P(C)} = \frac{P(A|C) \cdot P(B|C) \cdot P(C)}{P(B|C) \cdot P(C)}$$

$$\star \boxed{P(A|B, C) = P(A|C)} //$$

✱ when 'B' doesn't depend on A, we can remove the

* Bayesian Network



* It represents joint distribution using graph.

✱ It is Directed Acyclic Graph (DAG)

✱ Nodes: Random variable
Edges: Dependency.

Ex: Burglary Alarm (when someone stole in our home, this Alarm rings).

↳ Ring. (when)

1. Burglary (thiefs)

2. Earthquake (sometimes it rings)

* If we are not in home we don't hear the alarm, so we ask 2 neighbours. (if they hear alarm).

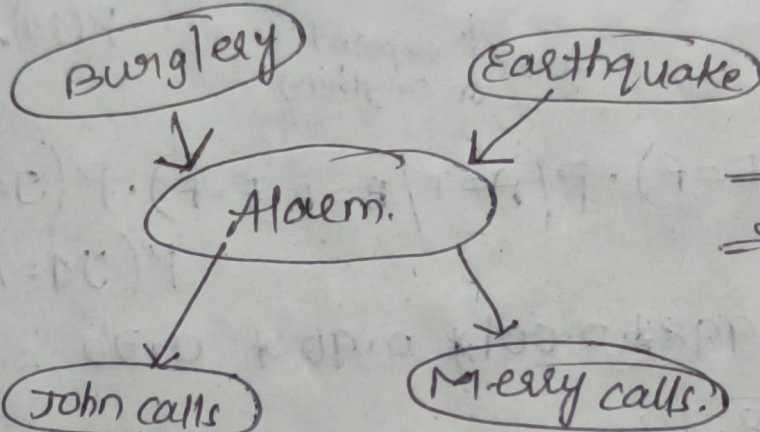
→ two neighbours

1. John } calm if they hear alarm ringing.
2. Melly } mistakes calls as alarm.
 } likes loud music

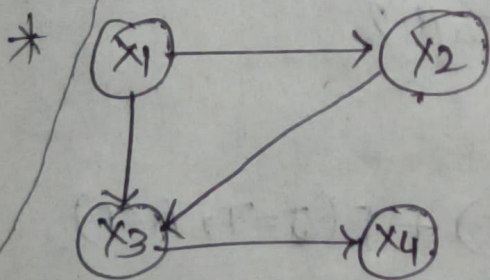
- * John is aged person, If phone is ringing, he said/ think alarm is ringing $\Rightarrow$ mistakes calls as alarm
- * Meery likes loud music $\Rightarrow$ If music is loud she doesn't hear alarm.

* We need to draw a graph.

Random Variables: $\Rightarrow$ Burglary, Earthquake, Alarm, John calls, Meery calls $\rightarrow [T/F]$



$\Rightarrow$ Design/graph.
 $\Rightarrow$ every node should contain probabilities



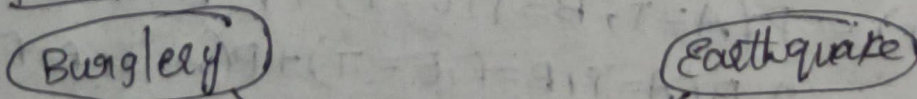
doesn't depend on anything
 x_2 depend on x_1
 x_3 depend on x_1, x_2

$$P(x_1, x_2, x_3, x_4) = P(x_1) * P(x_2/x_1) * P(x_3/x_1, x_2) * P(x_4/x_3)$$

It is called "factorization".

P(B)	
T	F
0.001	0.999

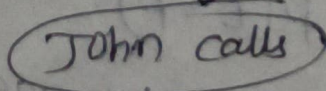
P(E)	
T	F
0.002	0.998



It is a complete Network.

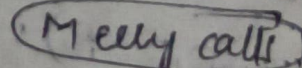
$P(A/B, E)$

B	E	T	F
T	T	0.95	0.05
T	F	0.99	0.06
F	T	0.29	0.71
F	F	0.001	0.999



$P(J/A)$

A	T	F
T	0.90	0.10
F	0.05	0.95



$P(M/A)$

A	T	F
T	0.70	0.30
F	0.01	0.99

collected from Data

* once the model is ready, we can Test it.
 (1) What is the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred & both John & Mary called?

Sol: $P(A=T, B=F, E=F, J=T, M=T) = ?$

$$P(B|E, A, J, M) = P(B) \cdot P(E) \cdot P(A|B, E) \cdot P(J|A) \cdot P(M|A)$$
 (Note: $P(B|E, A, J, M)$ doesn't depend on anything)

$$\therefore P(B=F) \cdot P(E=F) \cdot P(A=T|B=F, E=F) \cdot P(J=T|A=T) \cdot P(M=T|A=T)$$

$$= 0.999 * 0.998 * 0.001 * 0.90 * 0.70$$

$$= 0.000628$$

(2) What is the prob John calls?

Sol: $P(J=T) = ?$

$$= P(J=T|A) = P(J=T, A=T) + P(J=T, A=F)$$

$$\neq 0.90 + 0.70$$

$$P(J=T) \neq 0.97$$

$$P(J=T) = P(J=T|A=T) P(A=T) + P(J=T|A=F) P(A=F)$$

Now, $P(A=T) = P(A=T, B=T, E=T) + P(A=T, B=T, E=F) + P(A=T, B=F, E=T) + P(A=T, B=F, E=F)$

$$= P(A=T|B=T, E=T) \cdot P(B=T) \cdot P(E=T) +$$

$$P(A=T|B=T, E=F) \cdot P(B=T) \cdot P(E=F) +$$

$$P(A=T|B=F, E=T) \cdot P(B=F) \cdot P(E=T) +$$

$$P(A=T|B=F, E=F) \cdot P(B=F) \cdot P(E=F)$$

$$= 0.95 * 0.001 * 0.002 + 0.94 * 0.001 * 0.998 +$$

$$0.29 * 0.999 * 0.002 + 0.001 * 0.999 * 0.998$$

$$P(A=T) = 0.002516442 \approx 0.0025$$

$$\therefore P(A=F) = 1 - P(A=T)$$

$$= 1 - 0.0025$$

$$P(A=F) = 0.9975$$

$$\therefore P(J=T) = P(J=T/A=T) \cdot P(A=T) + P(J=T/A=F) \cdot P(A=F)$$

$$= 0.90 * 0.0025 + 0.05 * 0.9975$$

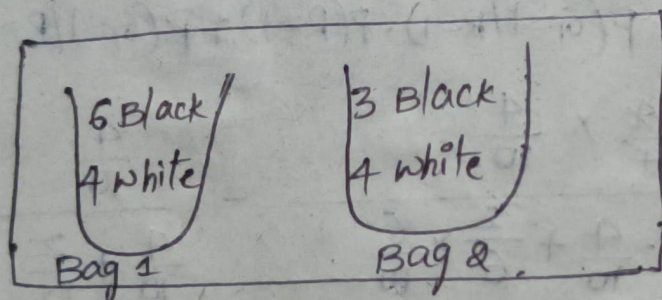
$$P(J=T) = 0.052125 //$$

$$\text{Probability of John calls} = 5.2125\% //$$

* Many Q's we are asking are in probabilistic Nature.

Continue

Ex: 1



$$P(\text{Black} | \text{Bag 1}) = \frac{6}{10} = 0.6$$

$$\therefore P(\text{Bag 1} | \text{Black}) = \frac{P(\text{Black} | \text{Bag 1}) \cdot P(\text{Bag 1})}{P(\text{Black})}$$

$$P(\text{Black} | \text{Bag 1}) = 6/10 = 0.6$$

$$P(\text{Bag 1}) = 1/2 = 0.5$$

$$P(\text{Black}) = P(\text{Black}, \text{Bag 1}) + P(\text{Black}, \text{Bag 2})$$

$$= P(\text{Black} | \text{Bag 1}) \cdot P(\text{Bag 1}) + P(\text{Black} | \text{Bag 2}) \cdot P(\text{Bag 2})$$

$$= \left(\frac{6}{10}\right)(0.5) + \left(\frac{3}{7}\right)(0.5)$$

$$= 0.51$$

$$\therefore P(\text{Bag 1} | \text{Black}) = \frac{0.6 * 0.5}{0.51} = 0.588 //$$

Continue
 (3) What is the prob. that alarm will ring?
 Sol: $P(A=T) = P(A=T, B=T, E=T) + P(A=T, B=T, E=F) + P(A=T, B=F, E=T) + P(A=T, B=F, E=F)$

$$\begin{aligned}
 &= P(A=T/B=T, E=T) \cdot P(B=T) \cdot P(E=T) + \\
 &P(A=T/B=T, E=F) \cdot P(B=T) \cdot P(E=F) + \\
 &P(A=T/B=F, E=T) \cdot P(B=F) \cdot P(E=T) + \\
 &P(A=T/B=F, E=F) \cdot P(B=F) \cdot P(E=F) \\
 &= 0.95 * 0.001 * 0.002 + 0.94 * 0.001 * 0.998 + \\
 &0.29 * 0.999 * 0.002 + 0.001 * 0.999 * 0.998
 \end{aligned}$$

$$= 0.0025$$

$$\therefore P(A=T) = 0.0025$$

$\therefore$ prob of Alarm will ring = 0.25%

9.e. $P(B \in A) = P(A|B \in E) \cdot P(B) \cdot P(E)$

$$P(A) = \sum_B \sum_E P(B \in A)$$

It has 4 comb's.

(4) What is the probability of alarm given John & Mary calls?

Sol: $P(A|J=T, M=T) = ?$

$$= P(A=T|M=T) = \frac{P(M=T|A=T) \cdot P(A=T)}{P(M=T)}$$

$$= \frac{0.70 * 0.0025}{P(M=T)}$$

$$P(M=T) = P(M=T|A) = P(M=T|A=T) + P(M=T|A=F)$$

$$= P(M=T|A=T) \cdot P(A=T) + P(M=T|A=F) \cdot P(A=F)$$

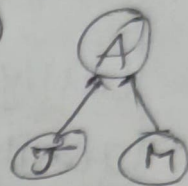
$$= 0.70 * 0.0025 + 0.01 * 0.9975$$

$$P(M=T) = 0.011725$$

$$P(A=T|M=T) = \frac{0.70 \times 0.0025}{0.011725}$$

$$P(A=T|J=T, M=T) = 0.149. \quad // \quad X$$

Sol:4 $P(A=T|J=T, M=T) = \frac{P(A=T, J=T, M=T)}{P(J=T|M=T)}$



~~$$P(A=T|J=T, M=T) = \frac{P(A=T, J=T, M=T)}{P(J=T|M=T)}$$~~

$$= \frac{P(A=T, J=T, M=T)}{P(A=T, J=T, M=T) + P(A=F, J=T, M=T)}$$

$$P(A=T, J=T, M=T) = P(J=T|A=T) \cdot P(M=T|A=T) \cdot P(A=T)$$

(b/c J, M are independent)

$$= 0.90 \times 0.70 \times 0.0025$$

$$= 0.001575$$

$$P(A=F, J=T, M=T) = P(J=T|A=F) \cdot P(M=T|A=F) \cdot P(A=F)$$

$$= 0.05 \times 0.01 \times 0.9975$$

$$= 0.00049875$$

$$\therefore P(A \neq T|J=T, M=T) = \frac{0.001575}{0.001575 + 0.00049875}$$

$$= 0.75949 //$$

* chain rule

$$P(X_1, X_2, X_3) = P(X_3|X_1, X_2) \cdot P(X_2|X_1) \cdot P(X_1)$$

$$P(X_1, \dots, X_n) = P(X_n|X_1, \dots, X_{n-1}) \cdot P(X_{n-1}|X_1, \dots, X_{n-2}) \dots P(X_1)$$

same as $= \prod P(X_i|X_1, \dots, X_{i-1})$

$$P(X_1, X_2, X_3) = P(X_1|X_2, X_3) \cdot P(X_2|X_3) \cdot P(X_3)$$

Ex 4 Effect — cavity (T/F)

Evidence — Toothache (T/F)

— probe catch (T/F)

∴ Table
 $2^3 = 8$ values

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	.108	.012	.092	.008
¬cavity	.016	.064	.444	.576

⇒ 8 values.

⇒ Joint distribution

$$\begin{aligned}
 \text{Ex 1 } P(\text{toothache}) &= P(\text{toothache} \wedge \text{catch} \wedge \text{cavity}) + \\
 &\quad P(\text{toothache} \wedge \neg \text{catch} \wedge \text{cavity}) + \\
 &\quad P(\text{toothache} \wedge \neg \text{catch} \wedge \neg \text{cavity}) + \\
 &\quad P(\text{toothache} \wedge \text{catch} \wedge \neg \text{cavity}) \\
 &= .108 + .012 + .016 + .064 \\
 &= \boxed{.20 = 20\%}
 \end{aligned}$$

$$\begin{aligned}
 P(\neg \text{catch}) &= P(\neg \text{catch} \wedge \text{cavity} \wedge \text{toothache}) + \\
 &\quad P(\neg \text{catch} \wedge \text{cavity} \wedge \neg \text{toothache}) + \\
 &\quad P(\neg \text{catch} \wedge \neg \text{cavity} \wedge \text{toothache}) + \\
 &\quad P(\neg \text{catch} \wedge \neg \text{cavity} \wedge \neg \text{toothache}) \\
 &= .012 + .064 + .008 + .576 \\
 &= \boxed{0.660} //
 \end{aligned}$$

$$P(\text{catch}) = 1 - 0.660 = \boxed{0.34} //$$

$$* P(\neg \text{cavity} | \text{toothache}) = \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

$$= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = \boxed{0.4}$$

$$* P(\text{cavity} | \neg \text{catch}) = \frac{P(\text{cavity} \wedge \neg \text{catch})}{P(\neg \text{catch})}$$

$$= \frac{0.012 + 0.008}{0.012 + 0.064 + 0.008 + 0.576}$$

$$P(\text{cavity} | \text{toothache, catch}) = \frac{P(\text{cavity} | \text{toothache, catch})}{P(\text{toothache, catch})}$$

$$= \frac{0.108}{0.108 + 0.016}$$

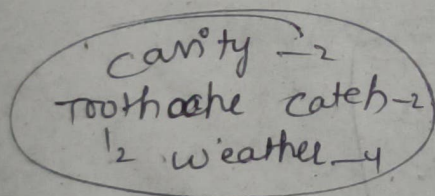
$$= \boxed{0.87} //$$

* for d-outcomes $\Rightarrow$ time-comp & s.c = $\boxed{d(d^n)}$ //

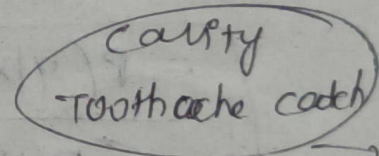
* Independence

$$P(A|B) = P(A) \quad , \quad P(B|A) = P(B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = \boxed{P(A)} //$$



decompose $\rightarrow$

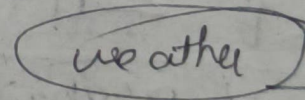


2.2.2 = 8

Total value = 2.2.2.4

= $\boxed{32}$ //

reduced $\rightarrow$



4

$\therefore$ total = 8 + 4 = $\boxed{12}$ //

* Bayes Theorem

$$P(x|y) = \frac{P(y|x) \cdot P(x)}{P(y)}$$

$$P(x, y) = P(x|y) \cdot P(y) = P(y|x) \cdot P(x)$$

$$* P(\text{cause} | \text{effect}) = \frac{P(\text{effect} | \text{cause}) \cdot P(\text{cause})}{P(\text{effect})}$$

* Diagnostic — finding cause for the problem/effect

$P(\text{car not running} | \text{no petrol})$ — causal

$P(\text{no petrol} | \text{car not running})$ — diagnostic

car not running — effect
no petrol — cause

$$\text{ex: } P(\text{toothache, catch} | \text{cavity})$$

$$= \frac{P(\text{toothache, catch, cavity})}{P(\text{cavity})}$$

$$= \frac{0.108}{0.108 + 0.012 + 0.072 + 0.008}$$

$$= \frac{0.108}{0.2} = \boxed{0.54} //$$

$$\star P(x|y) = \frac{P(y|x) \cdot P(x)}{P(y)}$$

$$\begin{aligned} \because P(y) &= P(y|x) + P(y|\neg x) \\ &= P(y|x) \cdot P(x) + \\ &\quad P(y|\neg x) \cdot P(\neg x) \end{aligned}$$

$$\boxed{P(x|y) = \frac{P(y|x) \cdot P(x)}{\sum P(y|x) \cdot P(x)}}$$

$$= \sum_x P(y|x) \cdot P(x)$$

$$\star P(x|y, z) = \frac{P(y|x, z) \cdot P(x|z)}{P(y|z)}$$

$$\boxed{= \frac{P(y|x, z) \cdot P(x|z)}{\sum P(y, x|z) \cdot P(x|z)}}$$

$$\boxed{= \alpha \cdot P(y|x, z) \cdot P(x|z)} //$$

$$\star P(\neg \text{cavity} | \text{toothache, catch}) =$$

$$\frac{P(\text{toothache, catch} | \neg \text{cavity}) \cdot P(\neg \text{cavity})}{P(\text{toothache, catch})}$$

$$= \frac{\alpha \cdot P(\text{toothache, catch} | \neg \text{cavity}) \cdot P(\neg \text{cavity})}{P(\text{toothache, catch})}$$

$$\text{ii) } P(\text{cavity} | \text{toothache, catch}) =$$

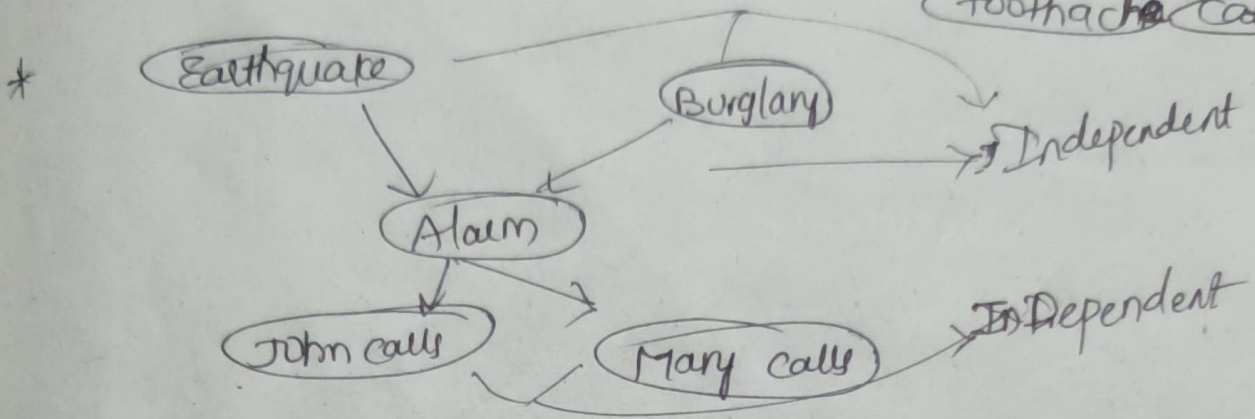
$$\frac{P(\text{toothache, catch} | \text{cavity}) \cdot P(\text{cavity})}{P(\text{toothache, catch})}$$

$$= \frac{\alpha \cdot P(\text{toothache, catch} | \text{cavity}) \cdot P(\text{cavity})}{P(\text{toothache, catch})}$$

$$\boxed{= \alpha \cdot P(\text{toothache, catch} | \text{cavity}) \cdot P(\text{cavity})}$$

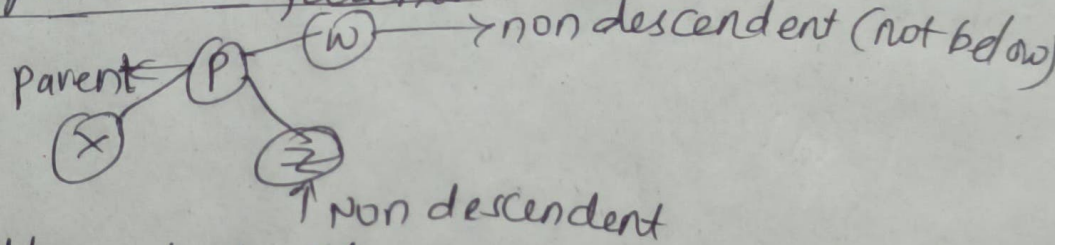
Now, we can compare α , if easily b/c denominator are all same.

* The dirⁿ is from cause to effect $\rightarrow$ cavity $\rightarrow$ toothache $\rightarrow$ Catch

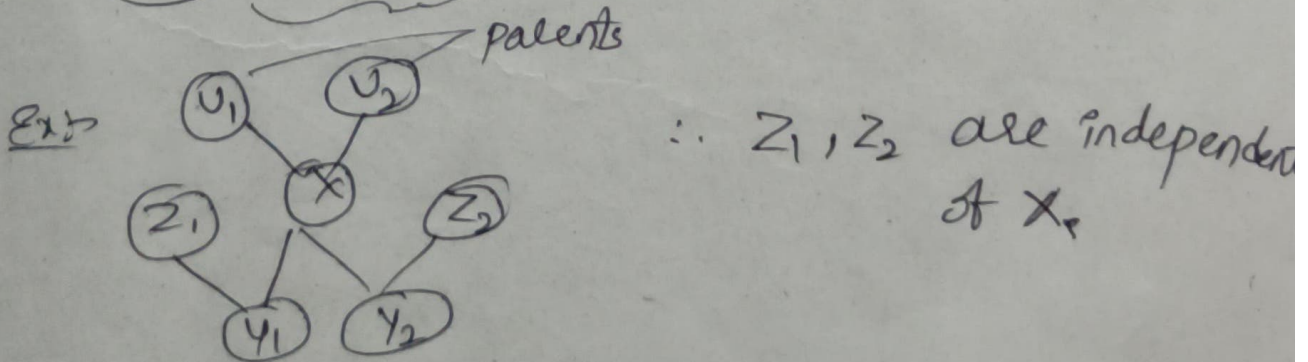
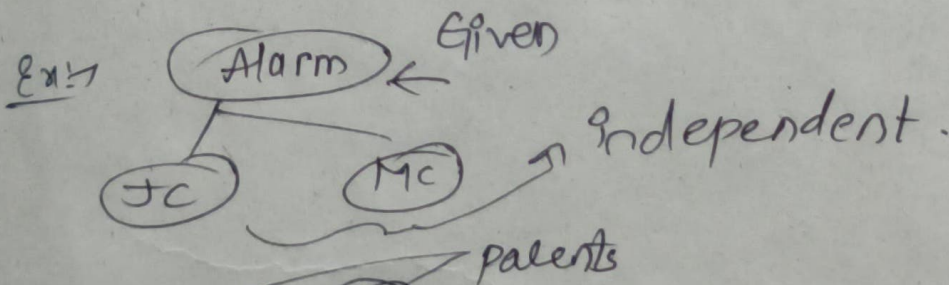


↓ If the o/p of 1 variable shouldn't change the prob of another $\Rightarrow$ Independent or else $\Rightarrow$ Dependent.

* 'X' is conditionally independent of all non descendants given its parents.



$\therefore X$ is independent of Z, W .



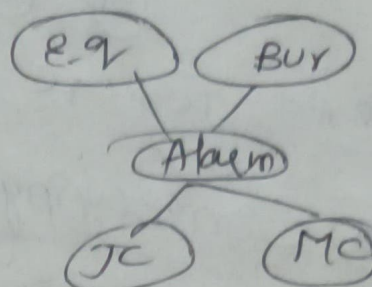
* Knowing about outcome of 1 variable doesn't affect the outcome of another variable - Independent or else - dependent.

$\rightarrow$ continue

$$P(JC, MC, A, E, B) = P(JC | MC, A, E, B) \cdot P(MC | A, E, B) \cdot P(A | E, B) \cdot P(E | B) \cdot P(B)$$

for $JC \rightarrow MC$, $Eg \rightarrow Bur$, $Bur \rightarrow Indep$

$MC \rightarrow$ parent is given, Eg , Bur all non-descendants $\rightarrow$ remove them



$$P(JC, MC, A, E, B) = P(JC | A) \cdot P(MC | A) \cdot P(A | B, E) \cdot P(E | B) \cdot P(B)$$

If parent is given then non-descendant nodes are independent to that node

Total $4 + 4 + 4 + 1 + 1 \Rightarrow$
 $\downarrow$
 (we can store 2 & find 2)
 $2 + 2 + 4 + 1 + 1 = 10$

$P(E | B) = E \text{ b/c}$
 $E | B \rightarrow \text{indep}$

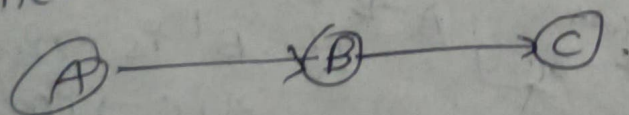
from 32 $\rightarrow$ we go to 10 parameters.

* Given parents, x is independent of non-descendants.

Given Markov Blanket, x is independent of All other nodes.

* d-separation (There are 3 rules).

① Linear connection:

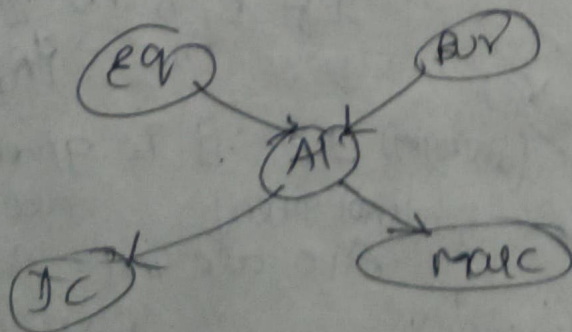


$\rightarrow B \text{ dep on } A$
 $C \text{ dep on } A$

* If 'B' is not given then info don't flow to C.

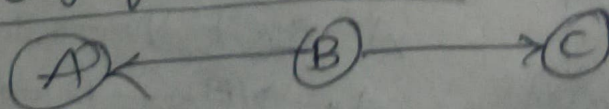
$\therefore$ If 'B' is given, $A \& C \Rightarrow \text{Independent}$.

Ex: 1

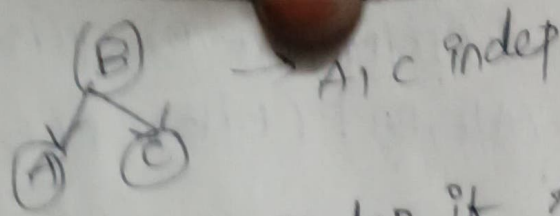


Album is given
 $Eg, MC \rightarrow \text{Indep}$
 $Eg, JC \rightarrow \text{Indep}$
 $Bur, JC \rightarrow \text{Indep}$
 $Bur, MC \rightarrow \text{Indep}$

② Diverging connections:



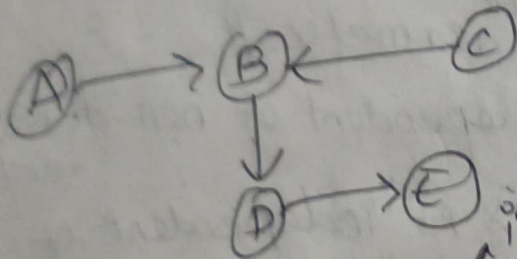
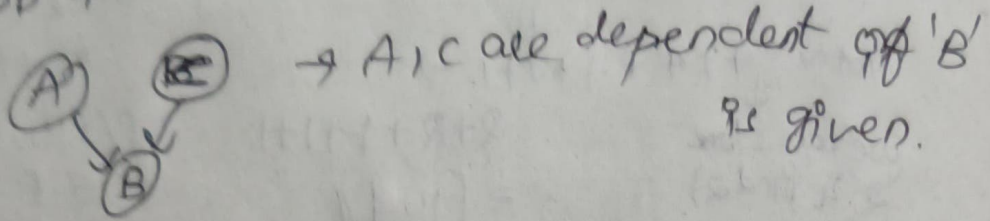
$\rightarrow B \text{ to } A$
 $B \text{ to } C$
 $\therefore B \text{ is given}$
 $A \& C \Rightarrow \text{Independent}$



Ex: J, M, C are indep if Alarm is given.

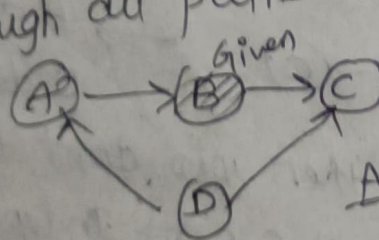
③ Converging connection

If Nothing is given A, C are indep (eg, Burg is indep)
If Alarm is given & eg happens, it decreases the prob of Burglary.



A, C are independent if B is not given.
If B, D, E all given (any descendant of B) then A, C are dependent

Ex: Through all paths they should be independent

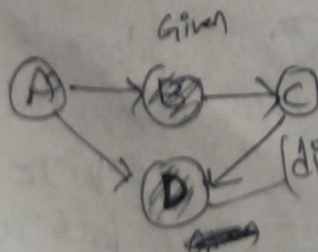


A-B-C
If B is given
A, C → indep.

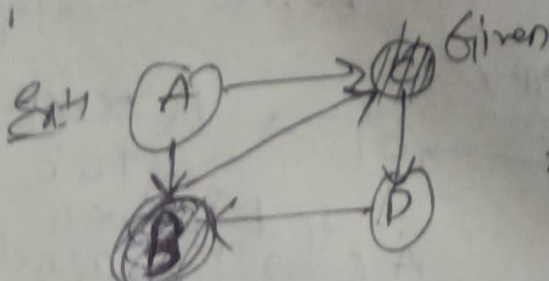
A-D-C

If D is not given
(we can't say anything)
A, C → dep. (we can't tell dep/indep)
If B, D are given then
A, C → independ.

Ex: 4



(diverging) If B is given A, C are indep.
& D is not given
A, C are ind A → B → C,
A → D → C path (converging)



A → C → D → indep
A → B ← D → depen

we can't say A and are Independent!

* $P(\text{dengue} / \text{body pain})$

hidden

Testable / observable

$\rightarrow P(\text{job} / \text{cgpa})$

$\rightarrow$ we are given observable & we need to find hidden variables.

* prob. calculation

$$\rightarrow P(B|j,m) = \frac{P(B,j,m)}{P(j,m)}$$

T, F. (j,m)

(B is capital)

(Capital both outcomes)

B $\rightarrow$ variable (T/F)

j $\rightarrow$ single value (T)

$$P(b|j,m) \rightarrow b=T, j=T, m=T$$

$$P(\neg b|j,m) \rightarrow b=F, j=T, m=T$$

$$P(B|j,m) \rightarrow B=T/F, j=T, j=F$$

$$P(B|j,m) = P(b|j,m) + P(\neg b|j,m)$$

$$P(B|j,m) = \frac{P(B,j,m)}{P(j,m)}$$

denominator $= \alpha \cdot P(B,j,m)$ — Always adds rem. variable

$$\text{Ex: } P(E) = \sum_{B,A,j,m} P(B,E,A,j,m) = \alpha \cdot \sum_{E,A} P(B,E,A,j,m) \rightarrow \text{Joint}$$

$$= \alpha \cdot \sum_{E,A} P(B) \cdot P(E) \cdot P(A|E,B) \cdot P(j|A)$$

write in topo-logical order. (top to bottom in graph)

E stops here

A stops here

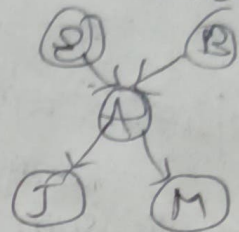
push sum until we get E, A.

$$= \alpha \cdot P(B) \cdot \sum_E P(E) \cdot \sum_A P(A|E,B) \cdot P(j|A) \cdot P(m|A)$$

$$\text{Ex: } P(j,m) = \sum_{B,E,A} P(B,E,A,j,m)$$

$$= \alpha \cdot \sum_{B,E,A} P(B) \cdot P(E) \cdot P(A|B,E) \cdot P(j|A) \cdot P(m|A)$$

$$P(j,m) = \alpha \cdot \sum_B P(B) \cdot \sum_E P(E) \cdot \sum_A P(A|B,E) \cdot P(j|A) \cdot P(m|A)$$



$$P(J) = \sum_{B, E, A, M} P(B|E, A, J, M)$$

$$\propto \sum_{B, E, A, M} P(B) \cdot P(E) \cdot P(A|B, E) \cdot P(J|A) \cdot P(M|A)$$

$$= \alpha \cdot \sum_B P(B) \cdot \sum_E P(E) \cdot \sum_A P(A|B, E) \cdot P(J|A) \cdot \sum_M P(M|A)$$

$$* P(b|j, m) = \alpha P(b) \cdot \sum_E P(E) \cdot \sum_A P(A|b, E) \cdot P(j|A) \cdot P(m|A)$$

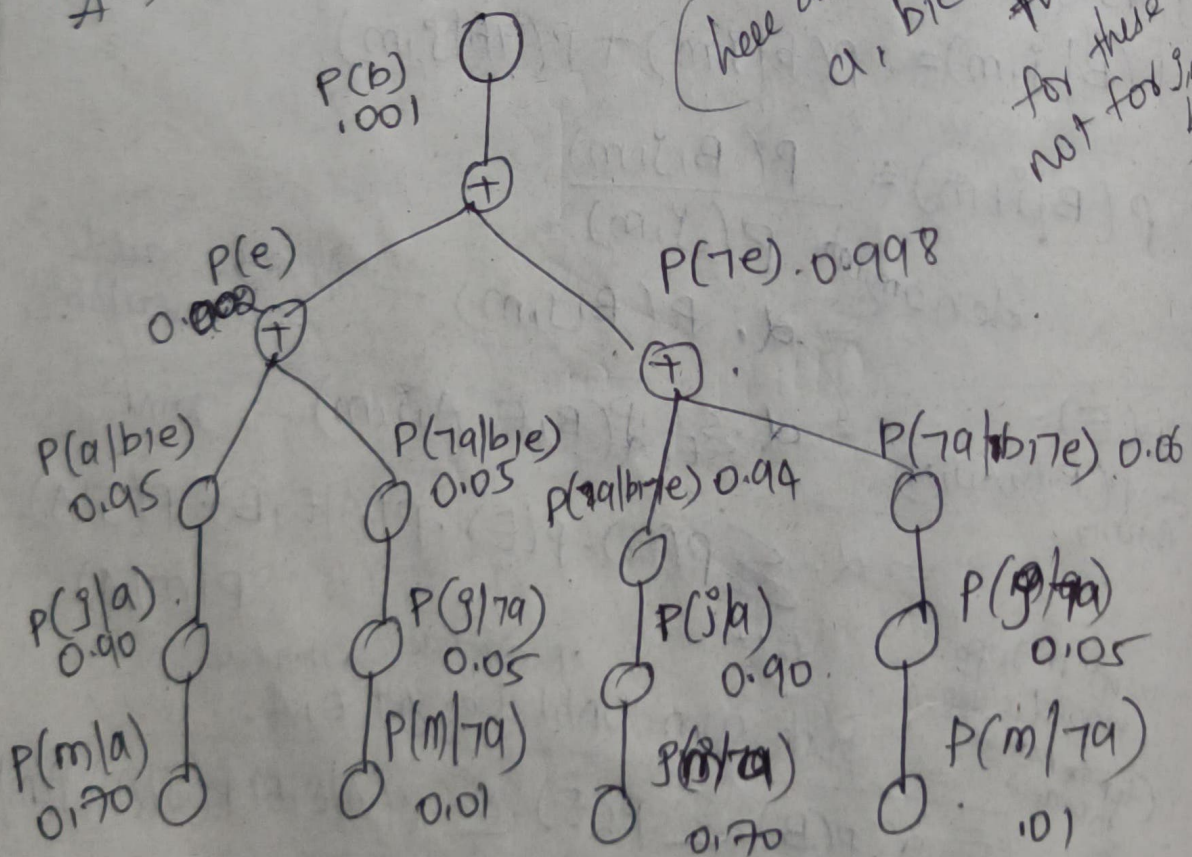
If there is sum $\rightarrow$ $\oplus$ node will come
 If there is multiplicⁿ $\rightarrow$ $\odot$ node will come.

$b \rightarrow$ only 1 value. $\left\{ \begin{array}{l} b.e + b.\neg e. \end{array} \right.$

$E \rightarrow e, \neg e.$

$A \rightarrow a, \neg a.$

(here also variables are a, b, e . only tests for these +ve & -ve not for j, m, b)



$$\begin{aligned} & \left[(0.70 \times 0.90 \times 0.95) + (0.01 \times 0.05 \times 0.05) \right] \times 0.02 \\ & + \left[(0.70 \times 0.90 \times 0.94) + (0.01 \times 0.05 \times 0.06) \right] \times 0.998 \end{aligned}$$

$$\therefore P(X=1) = \frac{0.026}{0.134} = \boxed{0.194} //$$

Ex:7 $\text{Rain} \rightarrow \text{Traffic} \rightarrow \text{Late}$

P	R(P)	R	T	P(T R)	T	L	P(L T)
T	0.1	T	T	0.8	T	T	0.3
F	0.9	T	F	0.2	T	F	0.7
		F	T	0.1	F	T	0.1
		F	F	0.9	F	F	0.9

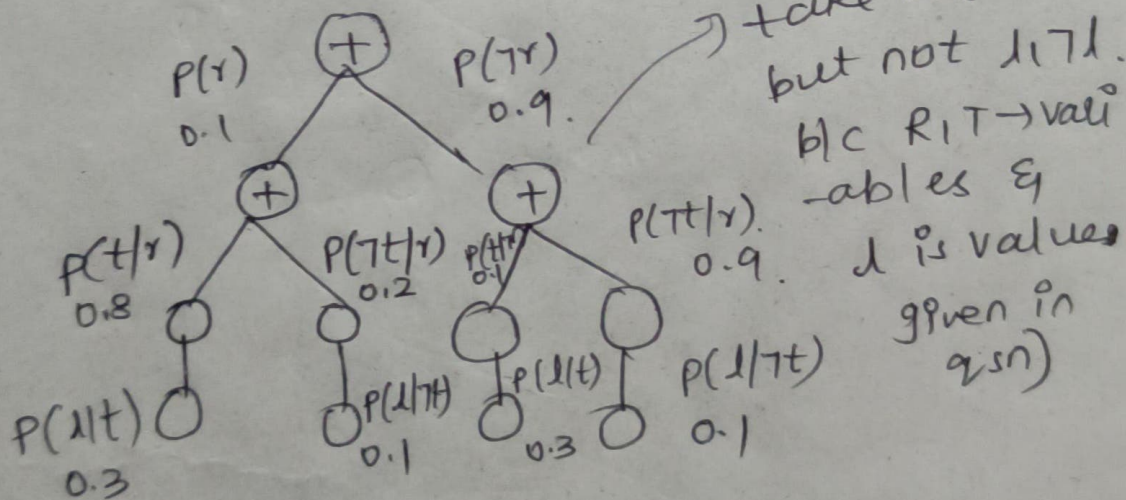
→ NO. of values stores
 $= 1 + 2 + 2 = 5$

$$P(L=T) = P(1) = ?$$

$$p(l) = \sum_{R, T} p(R, T, l)$$

$$= \sum_{R, T} P(R) \cdot P(T|R) \cdot P(1|T)$$

$$= \alpha \cdot \sum_R P(R) \cdot \sum_T P(T|R) \cdot P(1/T)$$



$$= [(0.3 * 0.8) + (0.1 * 0.2)] * 0.1 + \left[\left[\begin{matrix} (0.3 * 0.1) + (0.1 * 0.9) \\ * 0.9 \end{matrix} \right] \right]$$

$$= 0.026 + 0.108$$

$$P(1) = 0.134 //$$

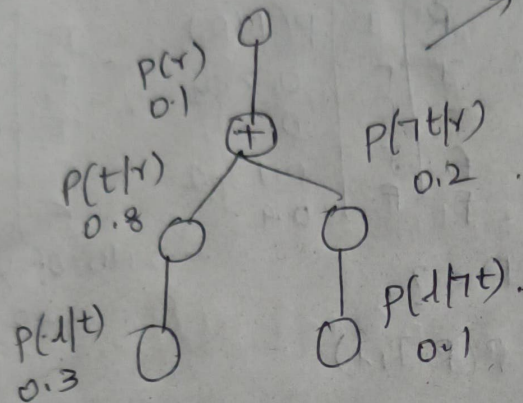
Ex 7 $P(8/1) = \frac{P(8,1)}{P(1)}$ ————— calculated.

$$= 2P(x, 1).$$

$$= \sum_T P(x_i | T)$$

$$= \sum_T P(Y) \cdot P(T/Y) \cdot P(1/T)$$

$$= P(Y) \cdot \sum_T P(T/Y) \cdot P(1/T)$$



(we need to take only t=7, but not 7=7 & 1=7 b/c 7=1 are values given in q. & T)

$$\rightarrow [(0.3 * 0.8) + (0.2 * 0.1)] * 0.1 = \boxed{0.026}$$

$$\therefore P(1/1) = \frac{0.026}{0.134} = \boxed{0.194} //$$